

Target validation paths — demo-nscclc-egfr-l858r-post-osu-d3m0

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

EGFR L858R, MET amplification (GCN 8.2), and T790M-absent are all confirmed at decision-relevant resolution, so no diagnostic test gates the savolitinib + osimertinib trial enrollment (SAFFRON / SACHI). The validation work here is about resistance characterization at progression, not target confirmation. The goal is to make sure the case really is MET-amplified bypass resistance and not something more complicated wearing the same imaging signature.

EGFR L858R

The most consequential miss right now would be SCLC histologic transformation, which is the highest-PPV resistance pattern that no liquid biopsy can confirm. Tumor NGS including TP53 and RB1 status is the screen: co-loss is the strongest predictor of transformation, with reported rates of 3–14% across post-osimertinib cohorts. If imaging morphology shifts or co-loss is detected, fresh re-biopsy with a neuroendocrine IHC panel (chromogranin, synaptophysin, INSM1, Ki-67) is the only way to confirm. ctDNA covering EGFR C797S and exon-20 inserts is the parallel on-target resistance check; a positive C797S result reframes the next-line conversation toward fourth-generation EGFR-TKIs or combination strategies.

MET amplification

The FISH GCN 8.2 result clears SAFFRON's enrollment threshold (≥ 6) cleanly, but MET amplification can be focal, and primary-vs-metastatic discordance is a documented confounder in this setting. A second-site FISH (or a matched archival block) refines confidence that the dominant disease, not just the biopsied site, is the amplified one. MET IHC (clone SP44) is the orthogonal-modality complement to the SAFFRON eligibility profile. A comprehensive NGS panel that picks up HER2 / BRAF / FGFR co-alterations frames durability expectations and the next-line plan.

Where to order these assays

The preferred provider for each assay is marked **(preferred)**, selected on company size, reputation, US-based location, and turnaround time. Other providers in the row are listed in case the preferred lab is unreachable for this patient.

Assay	Provider	Decision gated	Contact
ctDNA panel (EGFR C797S / T790M / exon-20)	Guardant Health (preferred) (Guardant360 CDx)	Distinguishes on-target (C797S, exon-20) vs bypass post-osimertinib resistance; informs fourth-gen TKI vs combination strategy	test info · 505 Penobscot Drive, Redwood City, CA 94063 · 1-855-698-8887
ctDNA panel	Foundation Medicine (FoundationOne Liquid CDx)	Distinguishes on-target vs bypass post-osimertinib resistance	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
ctDNA panel	Tempus (xF+)	Distinguishes on-target vs bypass post-osimertinib resistance	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137

ctDNA panel	Caris Life Sciences (Caris Assure)	Distinguishes on-target vs bypass post-osimertinib resistance	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
ctDNA panel	NeoGenomics Laboratories (NeoLAB Liquid)	Distinguishes on-target vs bypass post-osimertinib resistance	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Tumor NGS for TP53 / RB1 + bypass alterations	Foundation Medicine (preferred) (FoundationOne CDx)	Predicts SCLC histologic transformation; informs whether next-line stays adenocarcinoma-targeted or pivots to chemo	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Tumor NGS	Caris Life Sciences (Molecular Intelligence)	Predicts SCLC histologic transformation	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
Tumor NGS	Tempus (xT)	Predicts SCLC histologic transformation	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Tumor NGS	NeoGenomics Laboratories (NeoTYPE Comprehensive)	Predicts SCLC histologic transformation	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Tumor NGS	Memorial Sloan Kettering (MSK-IMPACT)	Predicts SCLC histologic transformation	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Re-biopsy + neuroendocrine IHC panel	Mayo Clinic Laboratories (preferred)	Confirms SCLC transformation when imaging shifts or TP53/RB1 co-loss is detected; pivots therapy entirely	test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710
Re-biopsy + neuroendocrine IHC	ARUP Laboratories	Confirms SCLC transformation when imaging shifts or TP53/RB1 co-loss is detected	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-242-2787
Re-biopsy + neuroendocrine IHC	LabCorp / Esoterix Oncology	Confirms SCLC transformation when imaging shifts or TP53/RB1 co-loss is detected	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363
Re-biopsy + neuroendocrine IHC	Quest Diagnostics	Confirms SCLC transformation when imaging shifts or TP53/RB1 co-loss is detected	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
Re-biopsy + neuroendocrine IHC	Memorial Sloan Kettering Diagnostic Molecular Pathology	Confirms SCLC transformation when imaging shifts or TP53/RB1 co-loss is detected	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
MET FISH (gene/CEP7 ratio)	NeoGenomics Laboratories (preferred)	Refines confidence in MET amp gating result for SAFFRON enrollment; does not gate eligibility	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
MET FISH	Foundation Medicine	Refines MET amp gating result; does not gate eligibility	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639

MET FISH	Mayo Clinic Laboratories	Refines MET amp gating result; does not gate eligibility	test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710
MET FISH	LabCorp / Esoterix Oncology	Refines MET amp gating result; does not gate eligibility	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363
MET FISH	Quest Diagnostics	Refines MET amp gating result; does not gate eligibility	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
MET IHC (clone SP44)	Mayo Clinic Laboratories (preferred)	Confirms MET protein-level expression; orthogonal complement to FISH but does not change SAFFRON eligibility	test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710
MET IHC	ARUP Laboratories	Confirms MET protein-level expression	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-242-2787
MET IHC	NeoGenomics Laboratories	Confirms MET protein-level expression	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
MET IHC	LabCorp / Esoterix Oncology	Confirms MET protein-level expression	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363
MET IHC	Quest Diagnostics	Confirms MET protein-level expression	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
Comprehensive bypass NGS (HER2 / BRAF / KRAS / FGFR)	Foundation Medicine (preferred) (FoundationOne CDx)	Frames durability expectations and the next-line plan for MET-directed therapy; does not gate SAFFRON	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Comprehensive bypass NGS	Caris Life Sciences (Molecular Intelligence)	Frames durability and next-line plan for MET-directed therapy	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
Comprehensive bypass NGS	Tempus (xT)	Frames durability and next-line plan for MET-directed therapy	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
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Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.